

Tech & Teach

Digital Chip Design Program

RTL Design

Mohammed alkhalifah

Mohammedalnaif

Task #05

8/7/2026

Combinational Datapath Design Part 1

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Task #1.1.1

```
module decoder_2to4 (
    input logic En,
    input logic w1, w0,
    output logic y0, y1, y2, y3
);
```

```
    always_comb begin
```

```
        y0 = 1'b0;
```

```
        y1 = 1'b0;
```

```
        y2 = 1'b0;
```

```
        y3 = 1'b0;
```

```
    if (En) begin
```

```
        case ({w1, w0})
```

```
            2'b00: y0 = 1'b1;
```

```
            2'b01: y1 = 1'b1;
```

```
            2'b10: y2 = 1'b1;
```

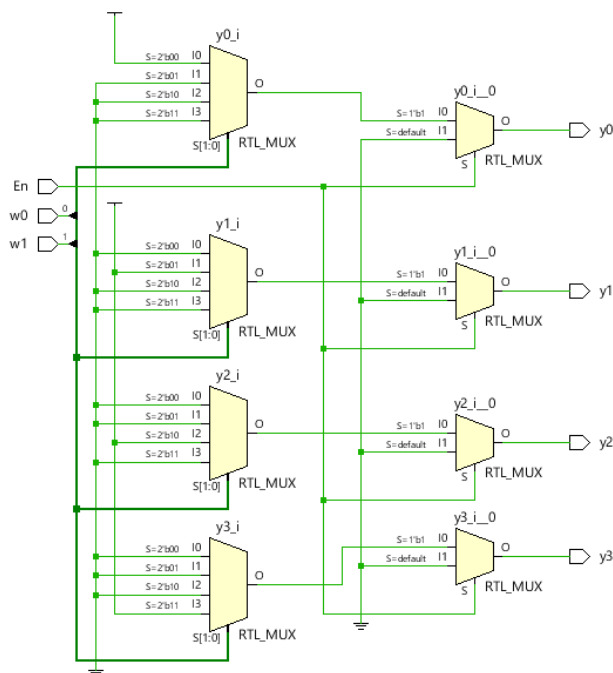
```
            2'b11: y3 = 1'b1;
```

```
        endcase
```

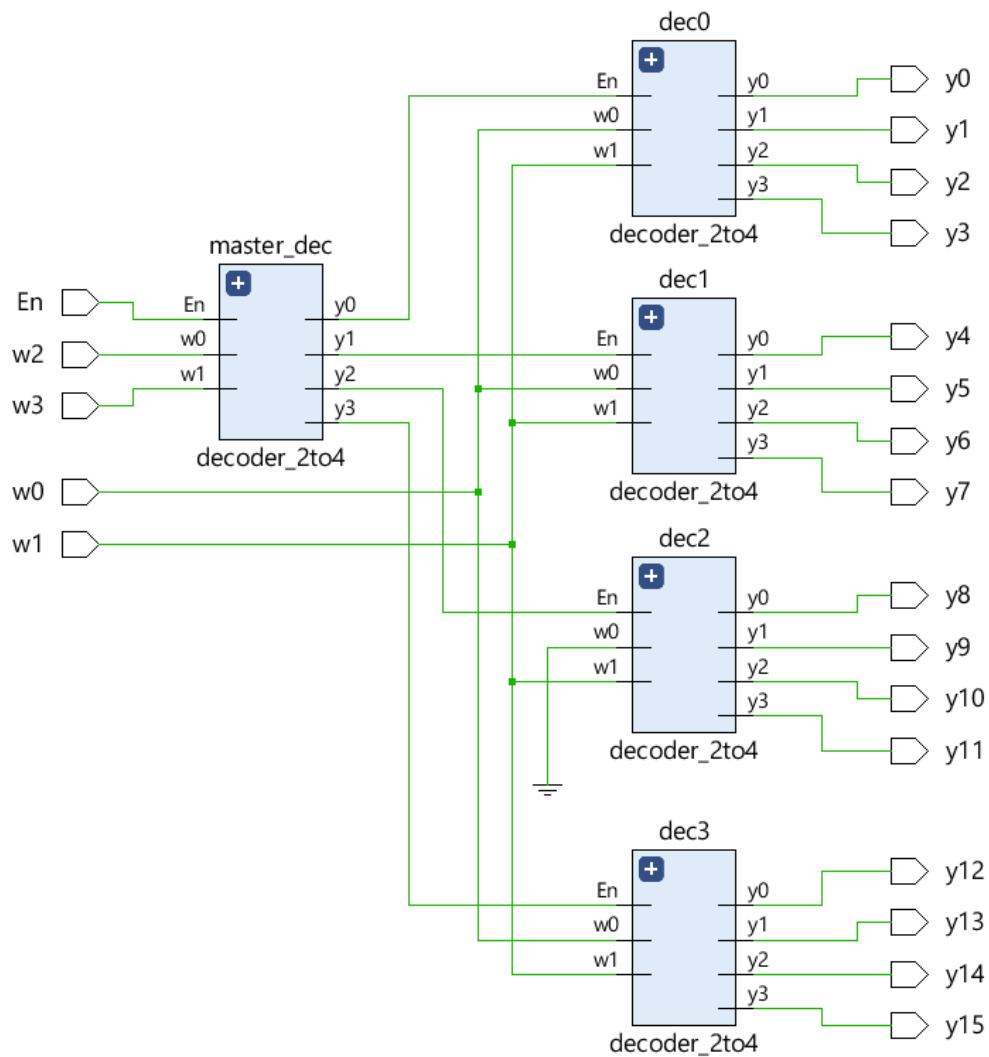
```
    end
```

```
end
```

```
endmodule
```

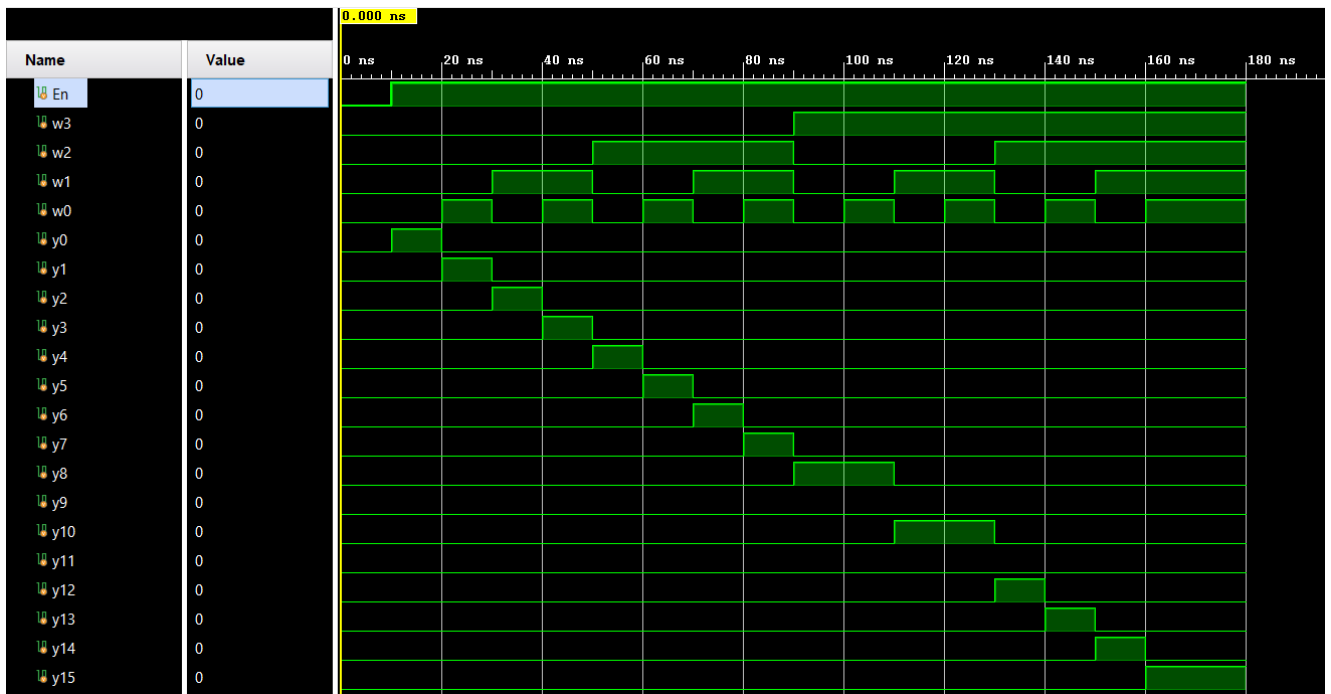


Task #1.1.2

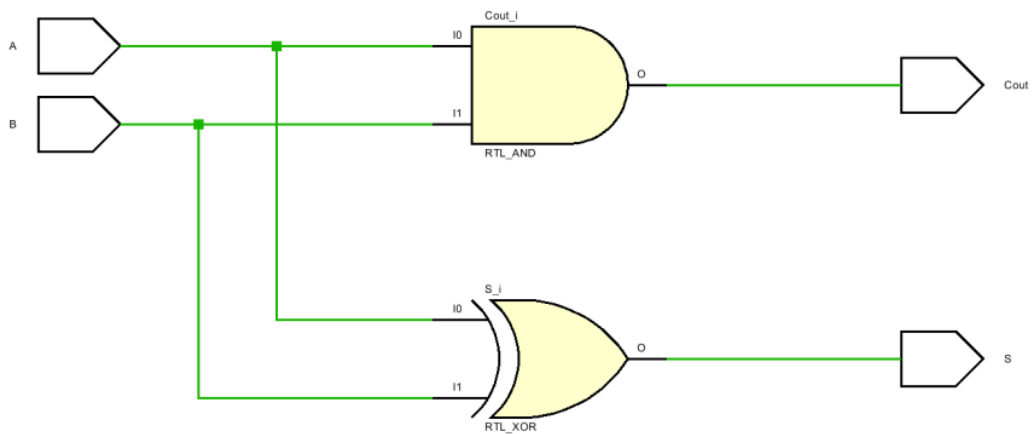


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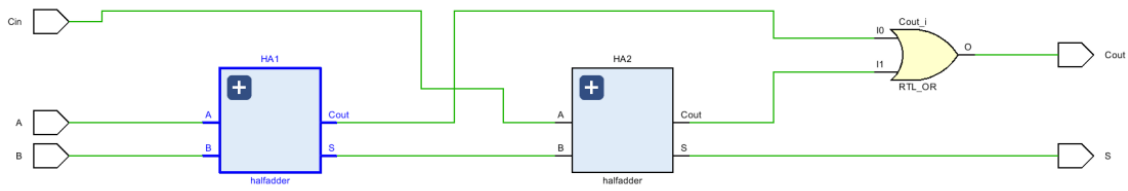
Task #1.1.3



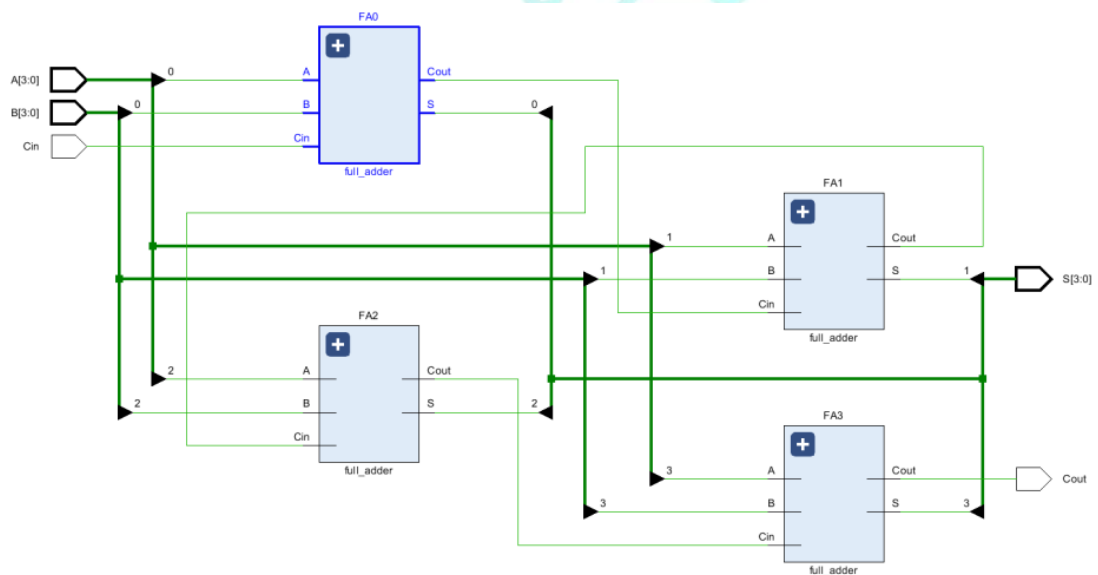
Task #1.2.1



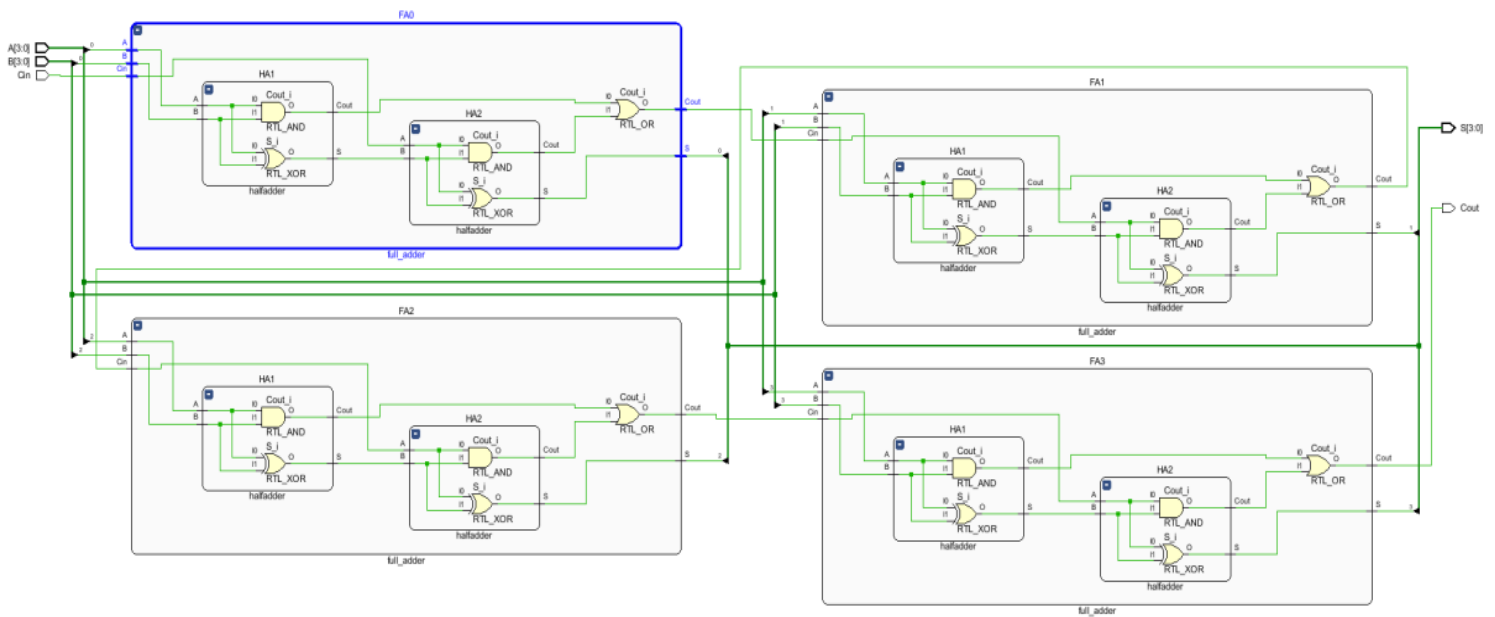
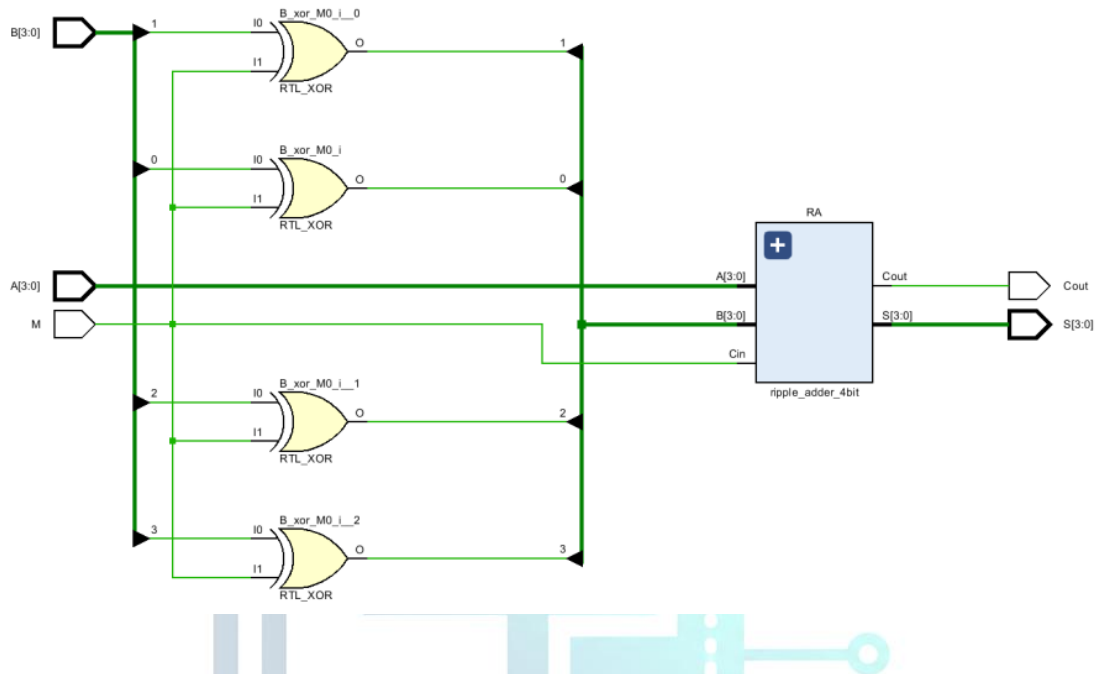
Task #1.2.2



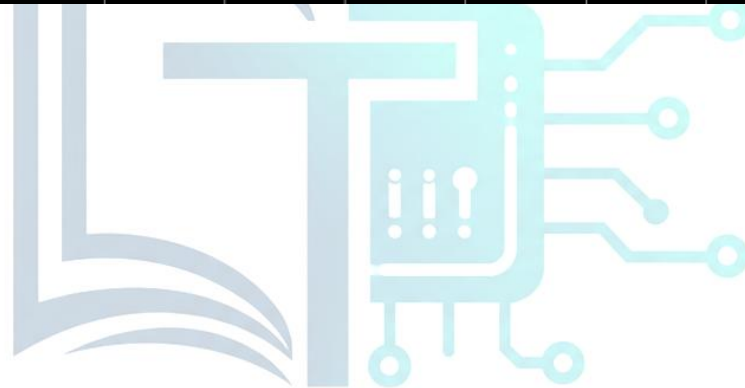
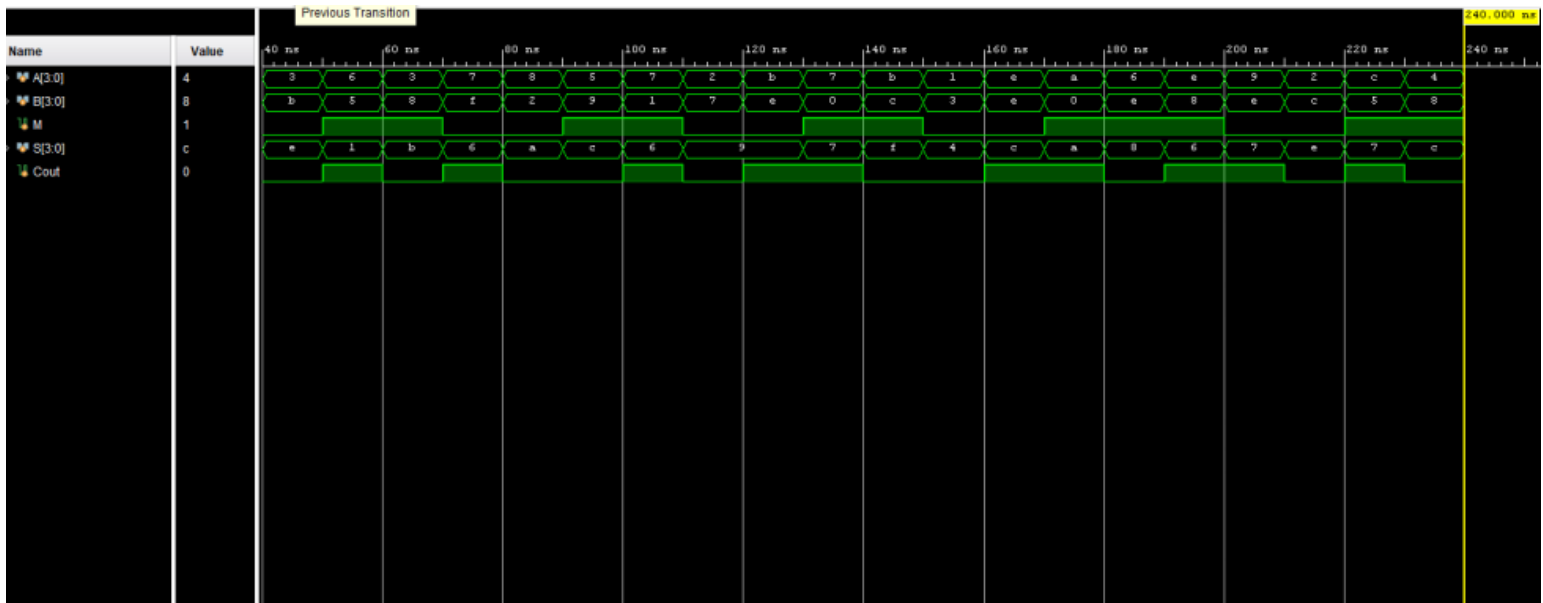
Task #1.2.3



Task #1.2.4

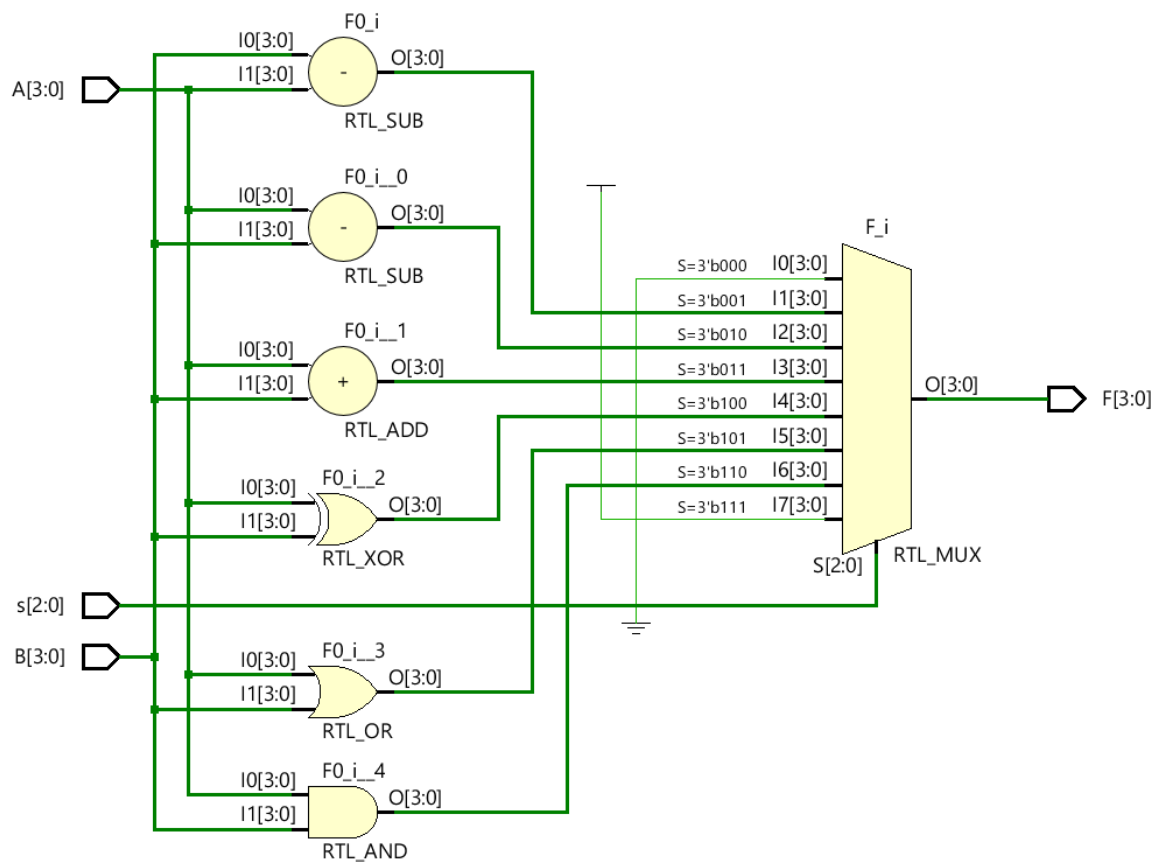
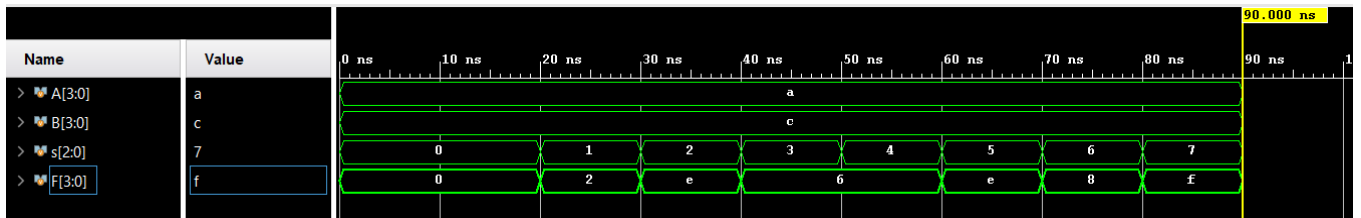


Task #1.2.5



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Task #3.1



Work distribution

Section	Abdulmalek	Omar	Khaled
Report	✓		
Task n.x		✓	
Task n.x+1			✓

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